# INDIAN ECONOMY

LEVEL II

## LECTURES

AN OVERVIEW ABOUT RICE AND WHEAT CROPS - Part I

## MODULE - 61

# KNOW THE FACTS ABOUT RICE

- Rice is primarily Kharif crop, accounting for around 85%.
- China is the largest rice producer at around 150 million tonnes annually. India comes second at around 125-130 million tonnes. However, the largest exporter is India.
- In 2020-21 and 2021-22, the value of the non-basmati exports are higher than the value of basmati rice exports. This happened after several years.
- In volumetric terms, basmati rice exports are around 4 million tonnes and non-basmati rice exports are around 17 million tonnes in 2021-22.
- India is the major player contributing to around 35-40% of global rice trade.

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# KNOW THE FACTS ABOUT RICE ... CONTD

- Between 1966 and 2011, though the total cropped area for Kharif Cereals remained nearly constant, harvested area dedicated to monsoon rice increased from 52% to 67%. Hence, we can conclude that the share increased from around half to two-third.
- Rice accounts for 74% of Kharif cereals production, 80% of energy and 81% of water used for cereal production in the season.
- Rice alone contributes to nearly 90% of Greenhouse Gas Emissions from Kharif Cereals.
- It is cultivated in around 45 million hectares in kharif and rabi seasons put together.
- In 2015-16, its production was 104 million tonnes and in 2021-22, it reached around 130 million tonnes.

## RICE ACCOUNTS FOR 40% OF TOTAL FOOD GRAINS!

- Out of the total food grains, Rice alone accounts for around 40%.
- Rice and Wheat put together constitutes around 70-75% of the total food grains.

# RICE PRODUCTION HAS SEEN CONSISTENT INCREASE SINCE 2015-16

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# RICE EXPORTS SKYROCKETED DURING 2020-21 AND 2021-22

# KNOW THE FACTS ABOUT WHEAT

- Wheat is exclusively Rabi crop, planted in around 30 million hectares.
- India is the world's second largest producer of wheat. China is the largest producer and Russia is the largest exporter.
- India's production was 92 million tonnes in 2015-16, and it is 105 million tonnes in 2021-22.
- Wheat is used not only for human consumption but also as poultry feed. The requirement for human consumption is around 91-93 million tonnes, whereas for poultry feed, the demand is around 4-5 million tonnes.
- Uttar Pradesh leads in production with close to one-third produced from the state. However, Punjab leads in productivity.
- It is incidental to point out that the average yield per hectare across the country is around 3 ½ tonnes per hectare.

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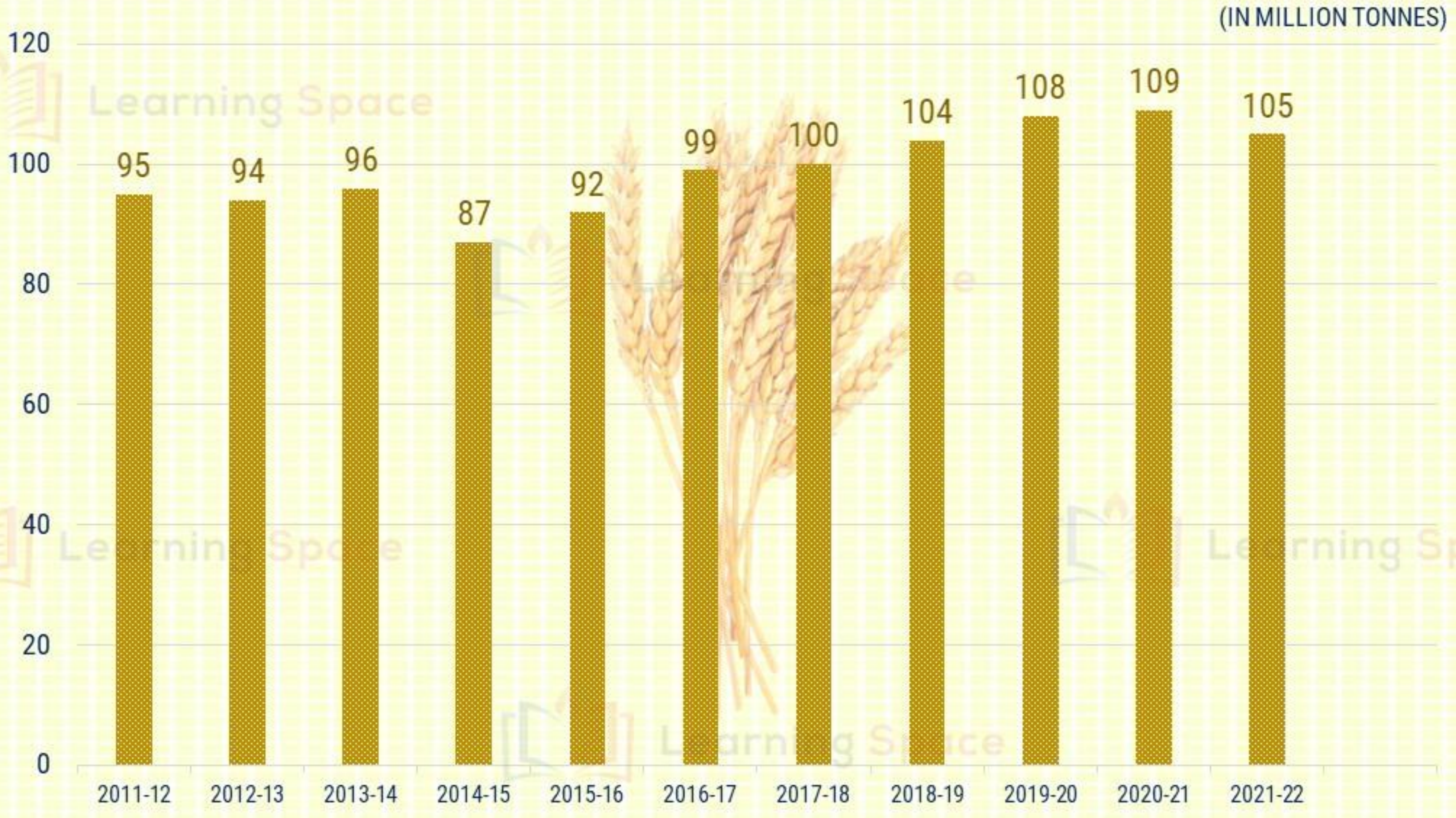
# WHEAT PRODUCTION DURING THE PREVIOUS DECADE ... DIP IN 2021-22



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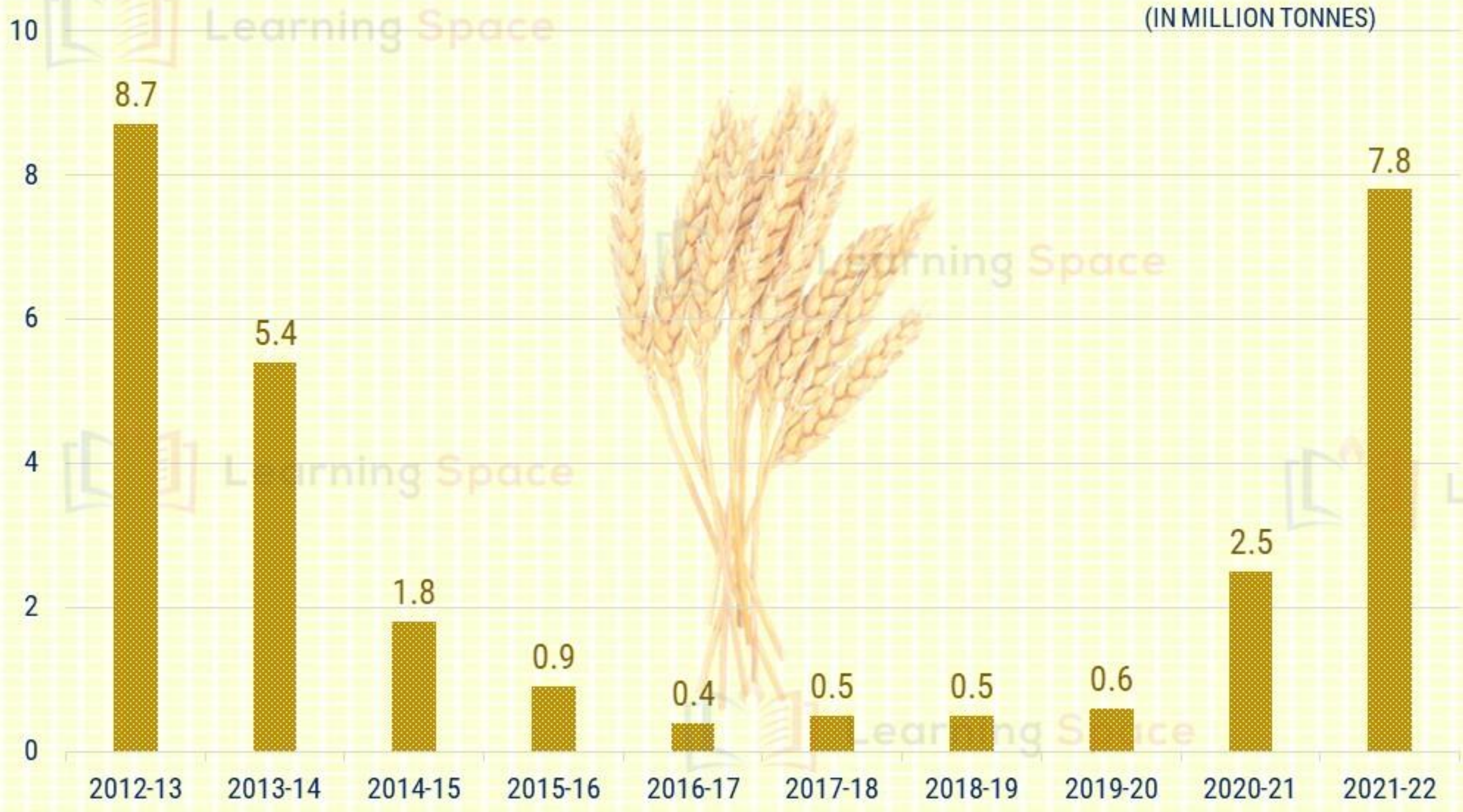
# WHEAT EXPORTS FROM INDIA SINCE 2012-13




***Global wheat prices rose nearly 50% within six months beginning with the start of the year 2022. Russia is the largest exporter of wheat and Ukraine is the sixth largest exporter, before the conflict.***

# WHAT ARE THE REASONS FOR THE ALMOST CONSISTENT INCREASE IN THE PRODUCTION OF RICE AND WHEAT?

- Annual hikes in the minimum support price combined with the system of open-ended procurement through the Food Corporation of India are the primary reasons.
- They have contributed not only to increase in harvest size but also for burgeoning public stocks of the two fine cereals.
- FCI often ends up buying 35% to 45% of the harvest.

भारतीय खाद्य निगम

Food Corporation of India

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# CULTIVATION OF RICE AND WHEAT IN SOME AREAS …

## … COMES WITH HUGE ENVIRONMENTAL COSTS

- Grain mono-cropping – cultivation of rice and wheat in an unbroken chain, season after season – in major growing States such as Punjab and Haryana over the last 20 to 30 years is inflicting enormous invisible costs.
- In the absence of scientific crop rotation, soil health has deteriorated.
- Encouraged by free/subsidised power supply, reckless drawing of groundwater resulted in the water table going down to alarmingly low levels.
- Environmental disaster is waiting to happen in the country's bread basket i.e., the north-western States.

## HOW DID THE RICE CULTIVATION START AROUND THE WORLD?

- Ever since rice cultivation started, farmers across the world followed the practice of broadcasting seeds directly on dry or puddled soils i.e., Direct Seeded Rice (DSR)
- The shift to Transplanted Rice (TPR) happened only in the past century because of two reasons:
  - ✓ The seed rate was very high, under DSR at around 100 kg per hectare, whereas transplanting rice needed just 10 to 15 kg per hectare.
  - ✓ Emergence of weeds in DSR and the lack of effective weedicides was another issue. In transplanting rice, the field is mostly waterlogged, weed growth is minimal.

## WHAT ABOUT INDIA?

- All the rice grown in the country a hundred years ago was through DSR. When Punjab and Haryana shifted from their traditional maize, millets, pulses and oilseeds to the wheat-paddy cultivation cycle, shift took place towards TPR.
- Subsequently, neither the Centre nor the States showed concern for the sustainability of the resources.

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## DIRECT-SEEDED RICE (DSR)

- It is also called Broadcasting Seed Technique.
- Globally, around 23% rice was directly seeded in 2007.

*As per Mexico-based International Maize and Wheat Improvement Centre, in India, about 10% of around 45 million hectares under rice cultivation is through DSR.*

## TRANSPLANTING RICE (TPR) SEEDLINGS

- It is the traditional and the prevalent way across India.
- Seedlings are developed from a nursery, then these are planted in the waterlogged fields.
- But, it is highly labour and water intensive.

# COST EFFECTIVENESS WITH DSR



## PUNJAB AGRICULTURAL UNIVERSITY REPORT SAYS SAVING OF Rs. 6,875 PER HECTARE!

### COST OF FIELD PREPARATION AND SOWING

Transplanting rice

**₹11,312 per ha**

Direct seeded rice

**₹5,312 per ha**

### YIELD

Transplanting

**8 tonnes per ha**

Direct seeded rice

**7.8 tonnes per ha**

### WATER REQUIRED TO PRODUCE 1 KG OF RICE

Transplanting

**3,000 litres**

Direct seeded rice

**2,100 litres**

### COST OF WEED AND NUTRIENT MANAGEMENT

Transplanting

**₹625 per ha**

Direct seeded rice

**₹2,450 per ha**

***Methane emissions were reduced by 6% to 92% by DSR method, depending on the type of variant used.***

# OVERALL, DSR HAS THE EDGE TOWARDS LONG-TERM SUSTAINABILITY!

## DEPLETION OF WATER TABLE CAN BE MINIMISED

- In States like Punjab, Haryana, Uttar Pradesh and Bihar, groundwater is the main source of irrigation.
- Rice cultivation by TPR has depleted the water table in several areas.
- In Punjab, the groundwater declined in about 85% of the State between 1984 and 2016. DSR can minimise the adverse effects on water table.

## BUT, WHAT ARE THE DRAWBACKS IN DSR?

- Rain, immediately after seeding, can reduce availability of soil nutrients and damage the crop.
- Germination can stop in certain areas of the field due to untimely rains.
- In spite of using weedicides, weeds may continue to be a problem.

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## WHAT ABOUT THE RODENT ATTACKS IN DSR FIELDS?

- DSR as a technique cannot be blamed for making the crop vulnerable to rodent attacks.
- In fact, several crops are grown with the Direct Seeding Technique.
- The failure is primarily because of the reason that rodent control campaign has not been launched before paddy sowing.

## PROBLEM OF RODENTS IS IGNORED TILL NOW DUE TO TPR!

- Rodents threaten every crop.
- Till now, as mostly paddy is grown by transplanting paddy seedlings in flooded fields, the menace of rats in rice is ignored.
- However, crop is vulnerable to attacks close to harvesting as rodents attack the roots.

## IN GENERAL, WHAT IS THE DAMAGE FROM RODENTS?

- In Punjab, on an average 2 to 15% damage is caused to every crop in Kharif and Rabi seasons by rodents.
- There is danger to every crop including wheat, paddy, pulses, vegetables etc.

## WHY RODENTS ARE A SERIOUS PROBLEM TO THE CROPS?

- They breed in vacant fields very fast after harvesting of any crop.
- They can reproduce up to 10 times or more per year. They have surprising reproductive capabilities and 4-8 weeks old mice are fully mature for reproduction.
- If they see that grains are enough to survive, they produce large number of offsprings. If there is a shortage of grain, then they produce in a controlled manner.

# HOW TO KEEP THE RODENTS UNDER CONTROL?



## NATURAL PREDATORS

- There are natural predators of rats, which include snakes, owls, mongoose, cats etc.
- Owl can eat 3-4 rats in one night.

## NARROW EARTHEN BUNDS / BOUNDARIES

- The bunds on the sides of the fields house several rats as they make their burrows here.
- Farmers should dismantle the bunds from time to time to restructure the new ones, which should be less raised and narrow.

***Weeds and grasses must also be cleared from around the fields, as they help in breeding of rodents.***

# WHAT ABOUT RODENTICIDES?

- Zinc phosphide and Bromadiolone are the most effective rodenticides to control the rodents.
- Both are Government controlled medicines and cannot be sold in open.
- They are very cheap and Government can supply these to the farmers easily and the cost is just Rs. 30 to 35 per hectare.



***The sustained campaign to control the rodents is missing, probably because of years of TPR. Now, with DSR, we have to look at the menace of rodents before sowing.***

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# IF INDIA WANTS TO CONTINUE RICE EXPORTS OF THIS SCALE … WE HAVE TO LOOK AT ALTERNATIVES!

- If we must continue with rice exports of this magnitude, the crop has to be farmed in a water-efficient manner and with a lower GHG footprint.
- Farming practices such as Alternate Wetting Drying (AWD), Direct-Seeded Rice (DSR) and micro-irrigation will have to be taken up on a war footing.
- Farmers may be incentivized and rewarded to save water.
- At the same time, the farmers may also be incentivized to switch to less water-guzzling crops, instead of rice and sugarcane.

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